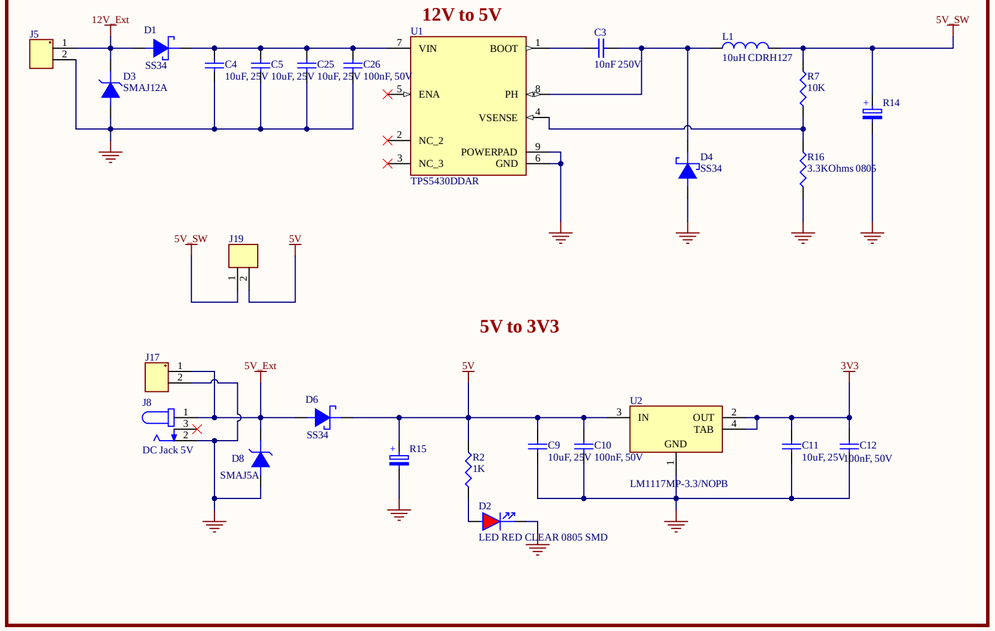
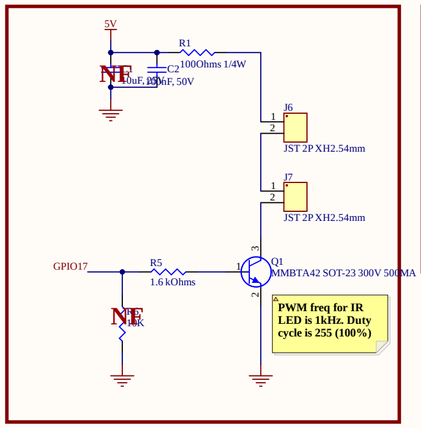
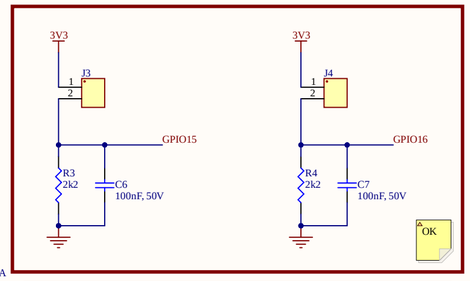
Power Supply



IR Emitters

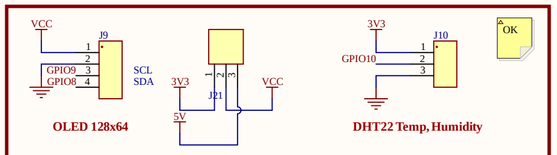


Phototransistors

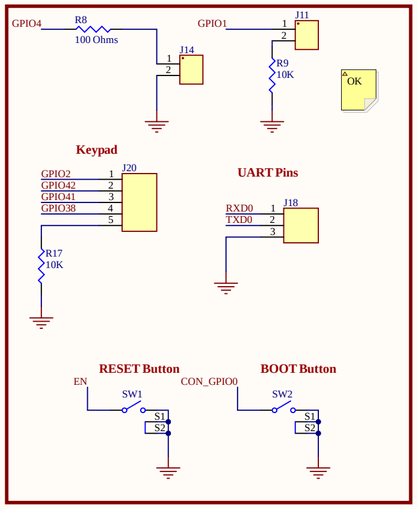


Connecting a 100nF cap to the ADC reading pin in use to mitigate the noise

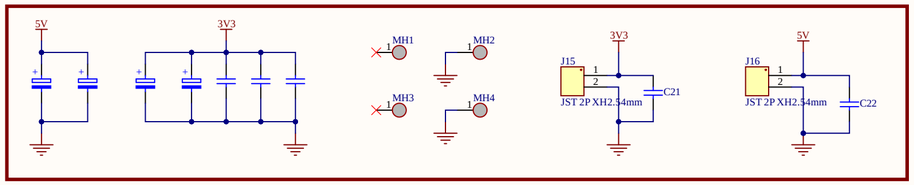
OLED and DHT22



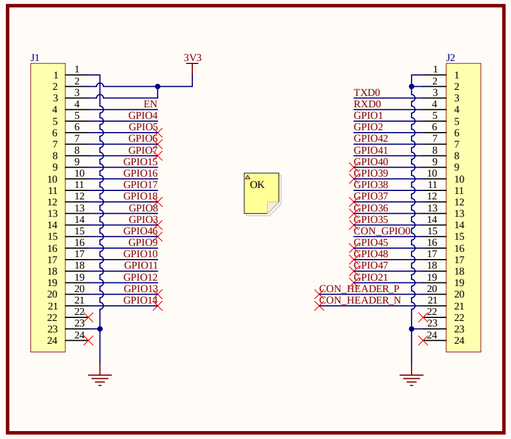
Indicator LED & UART Pins & Buttons



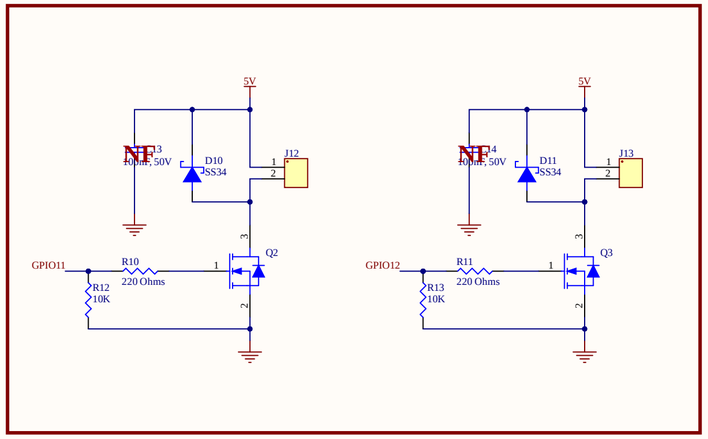
Bulk & Decoupling Caps & Mouting Holes & Ouput Pins



Header to ESP32



Pump Controller



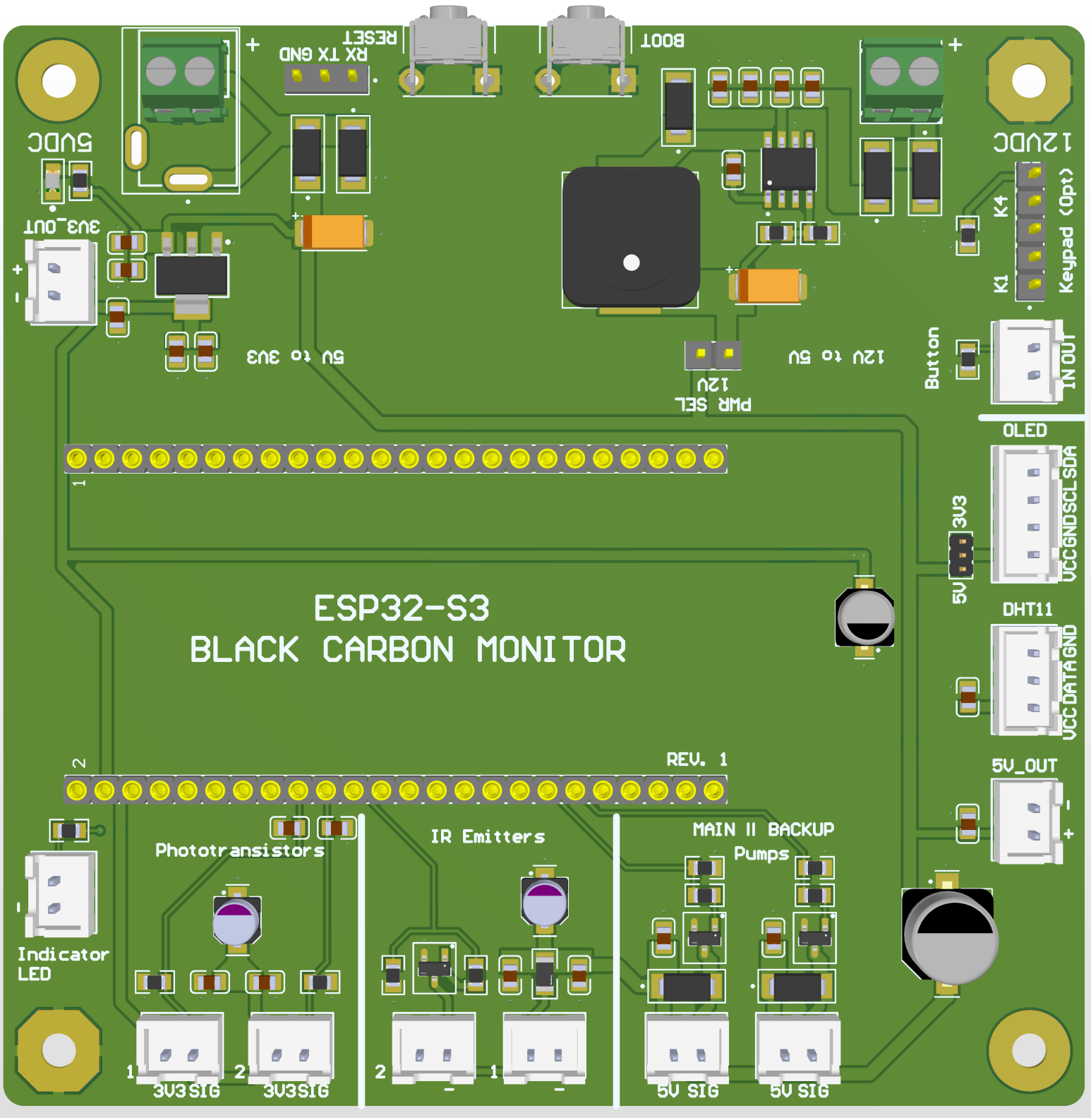
GPIOs
IR Emitters 17
Phototransistors 15 16
Pumps (main) (aux) 11 12
DHT22 10
OLED 9 8
Keypad 2 42 41 38
LED 4
Interrupt Button 1

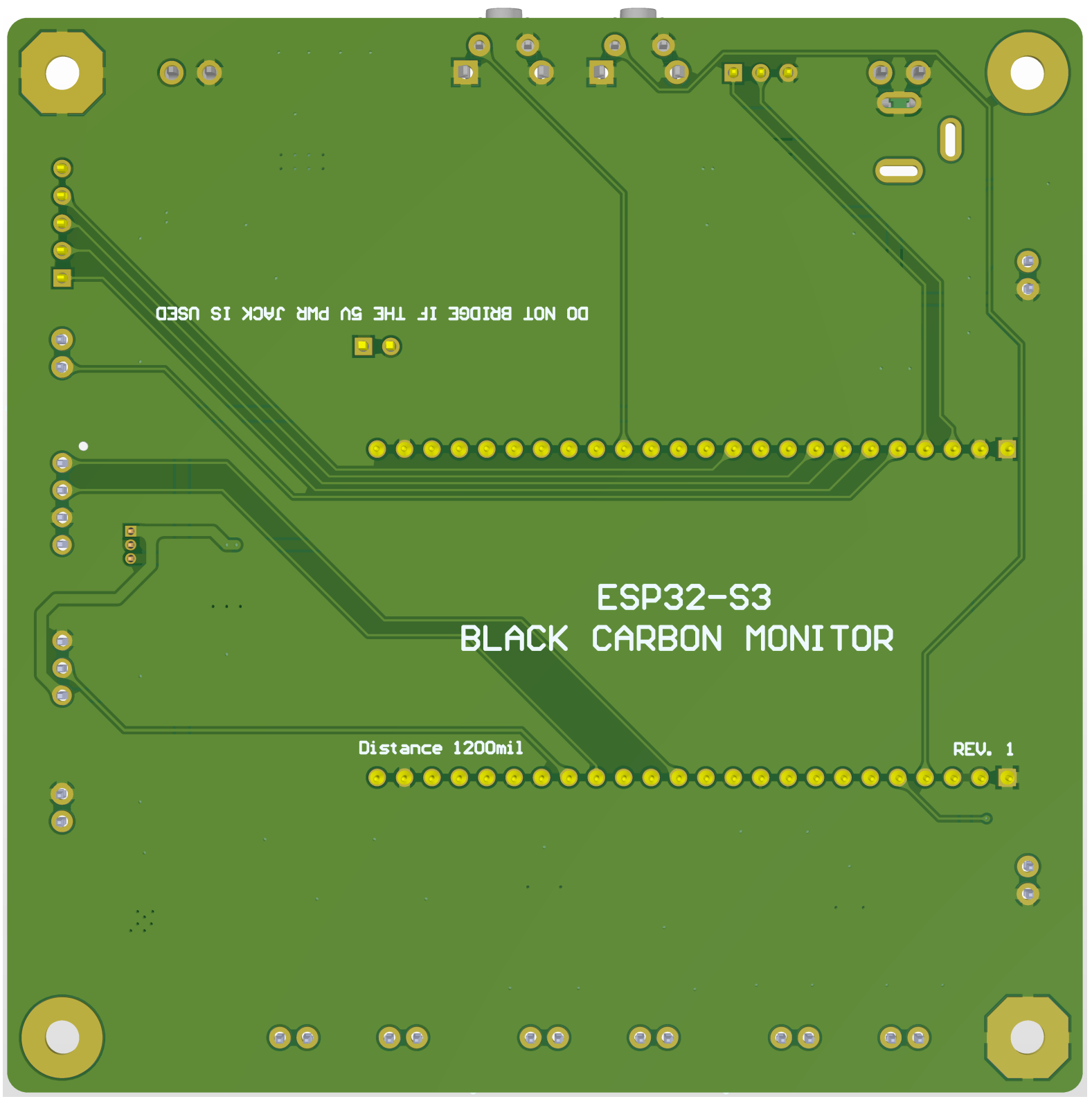
Strapping Pins

0	Pull-Up
3	Floating
45	Pull-Down
46	Pull-Down

Adjust the value of the resistor to change the sensitivity. The higher the resistance, the higher the sensitivity. The lower the resistance, the lower the sensitivity

Title		
Size: A2	Number	Revision
Date: 10/30/2026		Sheet of
File: E:\PCB_Sensor\Sensors_Board.SchDoc		Drawn By:


ESP32-S3
BLACK CARBON MONITOR
REV. 1
5VDC
12VDC
3V3_OUT
RX TX GND
RESET
BOOT
5V to 3V3
12V to 5V
PWR SEL
12V
Button
IN OUT
Keypad (Opt)
K4
K1
OLED
VCC GND SCL SDA
3V3
5V
DHT11
VCC DATA GND
5V_OUT
Indicator LED
Phototransistors
IR Emitters
MAIN II BACKUP
Pumps
3V3 SIG
3V3 SIG
5V SIG
5V SIG


DO NOT BRIDGE IF THE 5V PWR JACK IS USED
ESP32-S3
BLACK CARBON MONITOR
Distance 1200mil
REV. 1